# EEE4022S Rev0 PCB Motor Procurement Checklist and Assembly Guide

**Olwethu Khanzi | AK 07 PCB Motor Project**

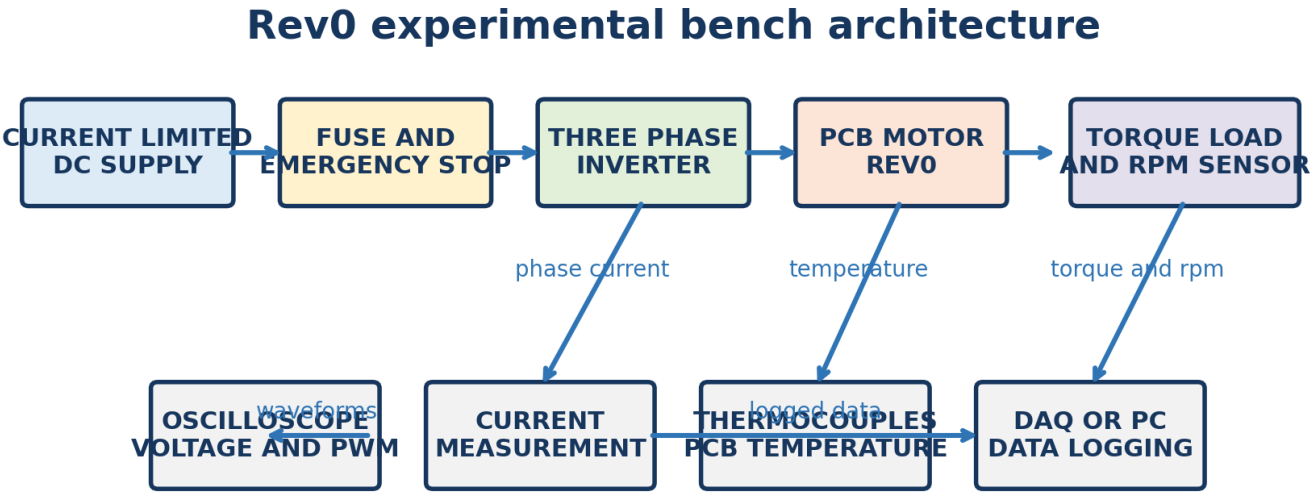



*Figure 1 Rev0 motor drive measurement and safety architecture*

**Purpose** This pack provides a request-ready checklist for the mechanical motor, drive electronics, measurement equipment and safety system needed to assemble and test the single-rotor five-PCB Rev0 machine. Each entry includes a representative engineering picture, quantity, specification, wording that can be sent to a technician or supplier, and a possible source.

**Important release condition** The 28 mm magnet-centre radius and 2 mm rotor back-iron are baseline dimensions. Release the rotor steel and final holder for manufacture only after the planned Maxwell sweeps of magnet radius 27, 28 and 29 mm and back-iron thickness 1.5, 2 and 3 mm.

## How to use this checklist

| | |
|---|---|
| **ORDERED** | No action unless the delivered part differs from the specification. |
| **FABRICATE** | Send the stated request and final drawing to the UCT workshop or 3D-printing facility. |
| **BUY** | Request a quotation or purchase the standard item from a local supplier. |
| **LAB** | Reserve or borrow from the UCT laboratory before spending project funds. |

### Locked Rev0 interfaces

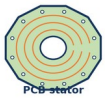
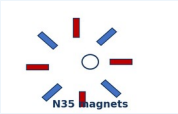
| Interface | Locked baseline |
|---|---|
| **PCB stack** | 5 x 1.0 mm PCBs; nominal stack approximately 5 mm |
| **PCB mounting** | 12 x 3.1 mm NPTH on 90 mm PCD; use six alternate holes |
| **Rotor magnets** | 16 x N35 magnets, 20 x 6 x 2 mm, alternating polarity |
| **Baseline magnet position** | 28.0 mm centre radius; 56.0 mm PCD |
| **Rotor** | 86 mm OD x 2 mm mild-steel baseline |
| **Shaft and bearings** | 8 mm shaft; 608-2RS bearings, 8 x 22 x 7 mm |
| **Initial air gap** | 1.0 mm magnet face to PCB surface; adjustable 0.5-2.0 mm |
| **Initial speed range** | Manual rotation first, then low-speed powered tests before higher-speed work |

### Controller warning

**Do not purchase a low-power gimbal-motor SimpleFOC shield merely because it is labelled BLDC.** The common DRV8313 SimpleFOC boards are intended for low current and relatively high winding resistance. This PCB motor is expected to have low resistance and low inductance. The inverter must therefore be selected only after phase resistance, inductance and Maxwell current estimates are known, with a current-limited bench supply used during commissioning.

## Parts already ordered

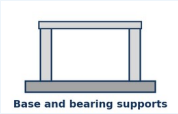
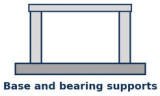
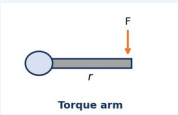
Verify these parts against the stated dimensions when they arrive and tick them off before starting the mechanical build.

| Tick | Picture | Item | Qty | Specification and request wording |
|---|---|---|---|---|
| ☐ |   | **PCB stator** **ORDERED** | **5** | **Specification** 12-sided board, 93 mm across flats, 25 mm centre hole, 1.0 mm thick, 6 copper layers, 12 x 3.1 mm NPTH on 90 mm PCD. **Request as** *Five bare Gould-design six-layer PCB stators manufactured at 1.0 mm thickness.* **Route** **Already ordered** \| JLCPCB |
| ☐ |   | **N35 magnets** **ORDERED** | **16 plus spares** | **Specification** Rectangular NdFeB, 20 x 6 x 2 mm, magnetised through the 2 mm thickness. Sixteen magnets per rotor face. **Request as** *N35 neodymium block magnets, 20 mm x 6 mm x 2 mm, axial magnetisation.* **Route** **Already ordered** \| UMAGNETS product |

## Parts to fabricate

These parts require a drawing, workshop operation or 3D print. The request wording can be copied directly into a workshop request.

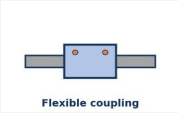
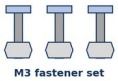
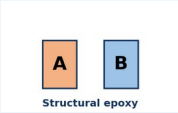
| Tick | Picture | Item | Qty | Specification and request wording |
|---|---|---|---|---|
| ☐ |  | **Rotor back iron**<br>FABRICATE | **1** | **Specification** Low-carbon mild-steel disc, baseline 86 mm OD x 2.0 mm, 18.1 mm centre pilot, 4 x 3.4 mm holes on 24 mm PCD.<br>**Request as** *Machine one mild-steel rotor disc to the supplied drawing. Do not use stainless steel.*<br>**Route** **UCT workshop** \| BMG engineering enquiry |
| ☐ |  | **Magnet holder**<br>FABRICATE | **1** | **Specification**<br>Nonmagnetic annulus, 84 mm OD, 30 mm ID, 2.1-2.2 mm thick, 16 radial pockets 20.2 x 6.2 mm on 56 mm PCD.<br>**Request as** *3D print one PETG or nylon magnet-retention plate plus a three-pocket tolerance coupon.*<br>**Route** **UCT 3D printing or local service** \| Micro Robotics PETG |
| ☐ |  | **Split clamp rotor hub**<br>FABRICATE | **1** | **Specification** 8 mm shaft bore, 32 mm flange OD, 18 mm locating pilot, 4 x M3 holes on 24 mm PCD, 15-20 mm boss.<br>**Request as** *Machine one split-clamp hub for an 8 mm shaft with an 18 mm rotor locating pilot.*<br>**Route** **UCT workshop** \| RS shaft hubs and collars |

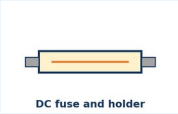
| Tick | Picture | Item | Qty | Specification and request wording |
|---|---|---|---|---|
| ☐ |  | **Base and bearing supports**<br>FABRICATE | **1 set** | **Specification** Rigid aluminium plate/profile preferred; two bearing blocks for 22 mm OD bearings; bearing spacing approximately 60-80 mm.<br>**Request as** *Fabricate a rigid base and two aligned 608 bearing supports from the final assembly drawing.*<br>**Route** **UCT workshop** \| UCT workshop |
| ☐ |  | **Stator clamp plates**<br>FABRICATE | **2** | **Specification**<br>Nonmagnetic perimeter clamps using six alternate 3.1 mm PCB holes on the 90 mm PCD; do not cover the active winding.<br>**Request as** *Fabricate front and rear stator perimeter clamps for a nominal 5 mm PCB stack.*<br>**Route** **UCT workshop or 3D print** \| UCT workshop |
| ☐ |  | **Torque arm**<br>FABRICATE | **1** | **Specification** Rigid aluminium arm with accurately measured radius, suggested 100 mm from shaft centre to load-cell force line.<br>**Request as** *Fabricate one rigid torque arm with a documented centre-to-force radius of 100.0 mm.*<br>**Route** **UCT workshop** \| UCT workshop |

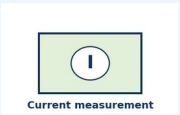
## Standard parts to buy

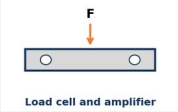
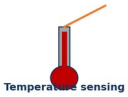
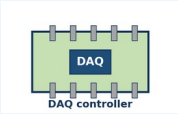
These are commercially available parts. A supplier link is included as a starting point, but equivalent parts are acceptable when every stated dimension and rating is met.

| Tick | Picture | Item | Qty | Specification and request wording |
|---|---|---|---|---|
| ☐ |   | **Precision shaft** BUY | **1 plus spare** | **Specification** 8 mm diameter precision-ground steel shaft, 250 mm initial length; h6 tolerance preferred. **Request as** *Supply an 8 mm diameter precision-ground steel shaft, approximately 250 mm long.* **Route Buy or workshop stock** \| RS South Africa |
| ☐ |   | **Sealed bearings** BUY | **2 plus 2 spares** | **Specification** 608-2RS deep-groove bearings, 8 mm bore x 22 mm OD x 7 mm width, rubber sealed both sides. **Request as** *Supply four 608-2RS sealed bearings, size 8 x 22 x 7 mm.* **Route Buy** \| BMG 608 bearings |
| ☐ |   | **Shaft collars** BUY | **4** | **Specification** 8 mm bore split-clamp collars preferred; avoid collars that permanently mark the shaft where possible. **Request as** *Supply four 8 mm bore one-piece or two-piece clamping shaft collars.* **Route Buy** \| RS shaft collars |

| Tick | Picture | Item | Qty | Specification and request wording |
|---|---|---|---|---|
| ☐ |  | **Flexible coupling**<br>BUY | **1** | **Specification** Motor side bore 8 mm; load-side bore selected after load or torque sensor shaft is known. Helical or jaw type.<br>**Request as** *Supply one flexible shaft coupling with an 8 mm motor-side bore; confirm the second bore before ordering.*<br>**Route** **Buy after load selected** \| Micro Robotics couplers |
| ☐ |  | **M3 fastener set**<br>BUY | **1 set** | **Specification** M3 x 8, 10 and 20 mm screws; M3 nyloc nuts; small OD steel and nylon washers; medium thread-lock.<br>**Request as** *Supply an M3 metric fastener assortment including six M3 x 20 screws and nylon PCB washers.*<br>**Route** **Buy** \| BMG fasteners |
| ☐ |  | **Structural epoxy**<br>BUY | **1** | **Specification** Two-part structural epoxy suitable for steel and nickel-coated magnets; temperature resistance above expected PCB temperature.<br>**Request as** *Supply a slow-cure two-part structural epoxy suitable for bonding NdFeB magnets to mild steel.*<br>**Route** **Buy** \| Pratley adhesives |

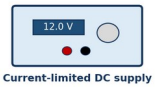
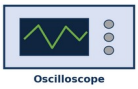
| Tick | Picture | Item | Qty | Specification and request wording |
|---|---|---|---|---|
| ☐ |  | **Feeler gauges and shims**<br>BUY | **1 set** | **Specification** 0.10, 0.20, 0.50 and 1.00 mm nonmagnetic shims or a metric feeler-gauge set.<br>**Request as** *Supply a metric feeler-gauge set covering 0.1 to 1.0 mm for air-gap adjustment.*<br>**Route Buy** \| RS measuring tools |
| ☐ |  | **Rotor safety guard**<br>BUY | **1** | **Specification** Clear 3 mm polycarbonate enclosure surrounding rotor and coupling; removable access panel; no acrylic near the rotor.<br>**Request as** *Supply a clear 3 mm polycarbonate sheet for an impact-resistant rotor guard.*<br>**Route Buy and fabricate** \| Maizey Plastics |
| ☐ |  | **Emergency stop**<br>BUY | **1** | **Specification** Latching mushroom emergency-stop switch controlling inverter power through a suitable relay or contactor.<br>**Request as** *Supply one normally-closed latching emergency-stop switch and appropriately rated switching device.*<br>**Route Buy or lab stock** \| RS emergency stops |
| ☐ |  | **DC fuse and holder**<br>BUY | **1 plus fuses** | **Specification** Inline DC fuse holder and several fuse values. Select final rating below cable and inverter limits after resistance tests.<br>**Request as** *Supply a DC fuse holder and a small range of suitable low-voltage DC fuses for current-limited commissioning.*<br>**Route Buy** \| Micro Robotics fuse holders |

| Tick | Picture | Item | Qty | Specification and request wording |
|---|---|---|---|---|
| ☐ |   | **Phase wiring and connectors** BUY | **1 set** | **Specification** Three colours of flexible copper wire, initially 16-18 AWG, ferrules, heat-shrink, insulated terminals and cable ties. **Request as** *Supply red, yellow and blue flexible 16-18 AWG wire plus ferrules, heat-shrink and insulated terminals.* **Route Buy** \| Micro Robotics wire |
| ☐ |   | **Current measurement** BUY | **1 DC plus phase access** | **Specification** Hall current sensor or calibrated shunt selected after expected current is confirmed; low insertion resistance preferred. **Request as** *Supply a bidirectional current sensor with a range comfortably above the expected motor current.* **Route Buy after Maxwell/current estimate** \| Allegro current sensor reference |
| ☐ |   | **RPM sensor** BUY | **1** | **Specification** Optical encoder, magnetic encoder or reflective tachometer; target at least one pulse per revolution and logged digital output. **Request as** *Supply a shaft-speed sensor compatible with an 8 mm shaft and capable of measuring 0-2000 rpm.* **Route Buy or lab stock** \| Micro Robotics encoders |

| Tick | Picture | Item | Qty | Specification and request wording |
|---|---|---|---|---|
| ☐ |  | **Load cell and amplifier**<br>BUY | **1 each** | **Specification** 5 kg straight-bar load cell and HX711-type amplifier for the torque arm; calibrate using known masses.<br>**Request as** *Supply one 5 kg load cell and one compatible 24-bit load-cell amplifier module.*<br>**Route Buy** \| Micro Robotics load-cell sensors |
| ☐ |  | **Temperature sensing**<br>BUY | **2 to 4** | **Specification** Fine K-type thermocouples, electrically insulated attachment, and MAX6675 or equivalent interface for logging.<br>**Request as** *Supply four fine K-type thermocouples and compatible amplifier/interface modules.*<br>**Route Buy** \| Micro Robotics thermocouples |
| ☐ |  | **DAQ controller**<br>BUY | **1** | **Specification** Arduino, STM32 or laboratory DAQ with sufficient analogue/digital channels for load cell, RPM, temperature and DC measurements.<br>**Request as** *Supply or reserve a data-acquisition controller and laptop connection for synchronized test logging.*<br>**Route Buy or borrow** \| Micro Robotics development boards |

## Laboratory equipment to reserve

Check UCT laboratory availability before purchasing. The inverter and instruments are more valuable as shared laboratory equipment than as once-off project purchases.

| Tick | Picture | Item | Qty | Specification and request wording |
|---|---|---|---|---|
| ☐ |  | **Current-limited DC supply**<br>LAB | 1 | **Specification** Bench supply initially 0-15 V, at least 10 A preferred, adjustable current limit, voltage and current displays.<br>**Request as** *Reserve a current-limited laboratory DC supply rated for at least 15 V and 10 A.*<br>**Route** **Borrow from UCT lab** \| Micro Robotics bench supplies |
| ☐ |  | **Three-phase inverter**<br>LAB | 1 | **Specification** 12-24 V capable, at least 10 A current capability, adjustable current limit, programmable PWM 20-40 kHz preferred, three phase outputs.<br>**Request as** *Provide a controllable low-voltage three-phase motor inverter suitable for a low-resistance, low-inductance PCB winding.*<br>**Route** **Borrow or build with supervisor approval** \| VESC platform reference |
| ☐ |  | **Oscilloscope**<br>LAB | 1 | **Specification** Four channels preferred, at least 50 MHz, differential probes or safe isolated voltage measurement, and current probe if available.<br>**Request as** *Reserve a four-channel oscilloscope and suitable probes for PWM, phase voltage and phase-current measurements.*<br>**Route** **Borrow from UCT lab** \| RS oscilloscopes |

| Tick | Picture | Item | Qty | Specification and request wording |
|---|---|---|---|---|
| ☐ |  | **Digital multimeter**<br>**LAB** | **2** | **Specification** Resistance resolution adequate to compare low phase resistances; voltage, current and continuity functions.<br>**Request as** *Reserve two digital multimeters, including one suitable for low-resistance comparison measurements.*<br>**Route Borrow from UCT lab** \| RS multimeters |

## Assembly sequence

Complete the assembly in the order below. Do not energise the inverter until the mechanical, electrical and safety checks in Steps 1 to 7 are complete.

### Step 1 Inspect the PCBs

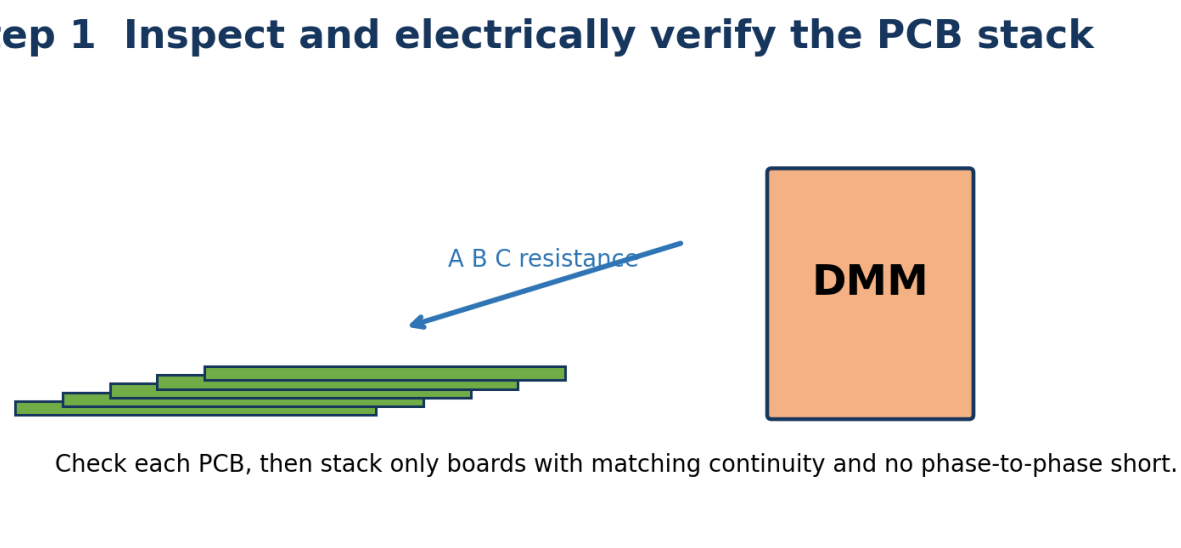



*Figure 2 Step 1 Inspect the PCBs*

Measure continuity and phase-to-phase resistance on each PCB. Record A-B, B-C and C-A resistance. Reject or investigate any board that differs materially from the others. Stack only electrically verified boards.

### Step 2 Build the magnet rotor

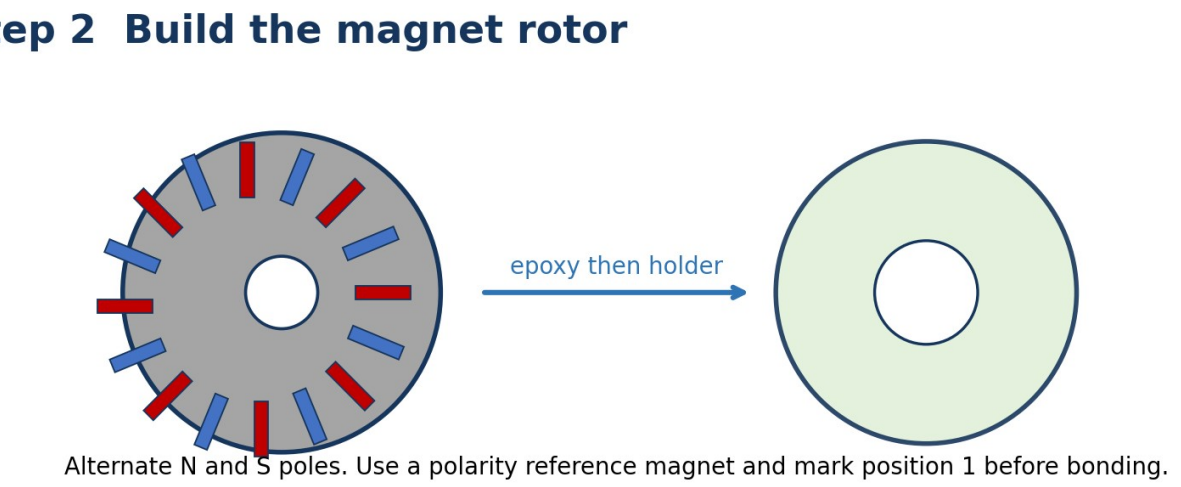



*Figure 3 Step 2 Build the magnet rotor*

Mark magnet position 1 and the required N-S sequence. Clean the mild-steel rotor and magnet surfaces. Dry-fit the tolerance-tested holder, bond one magnet at a time, and maintain alternating polarity. Allow the structural epoxy to cure fully.

### Step 3 Install shaft and bearings

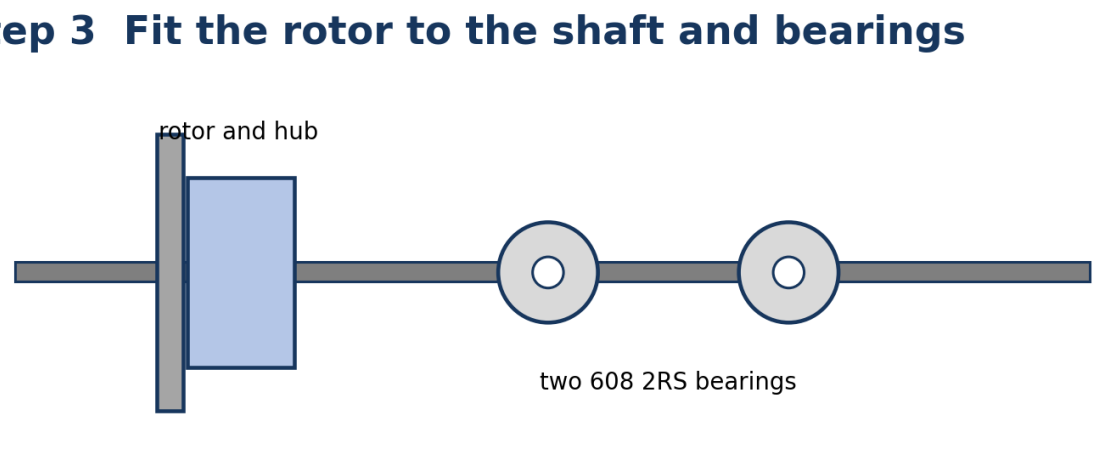



*Figure 4  Step 3 Install shaft and bearings*

Bolt the rotor disc to the centred hub, slide the hub onto the 8 mm shaft, and install both 608-2RS bearings and collars. Rotate by hand and check for eccentricity, axial wobble and bearing binding.

### Step 4 Clamp the stator stack

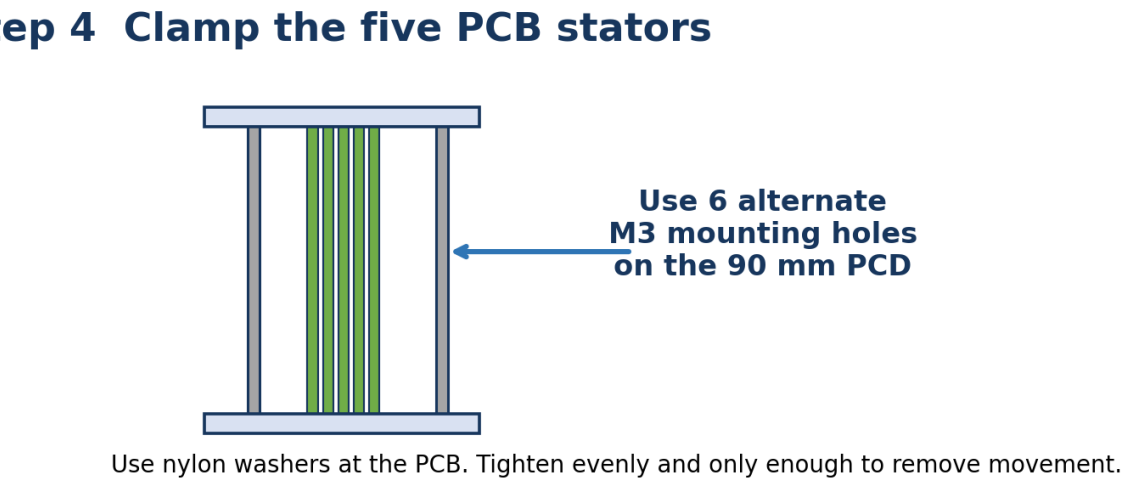



*Figure 5  Step 4 Clamp the stator stack*

Align the five PCB stators using six alternate 3.1 mm holes on the 90 mm PCD. Add nylon washers against the PCB and tighten the M3 fasteners gradually in a diagonal pattern. Do not crush the board edges.

### Step 5 Set alignment and air gap

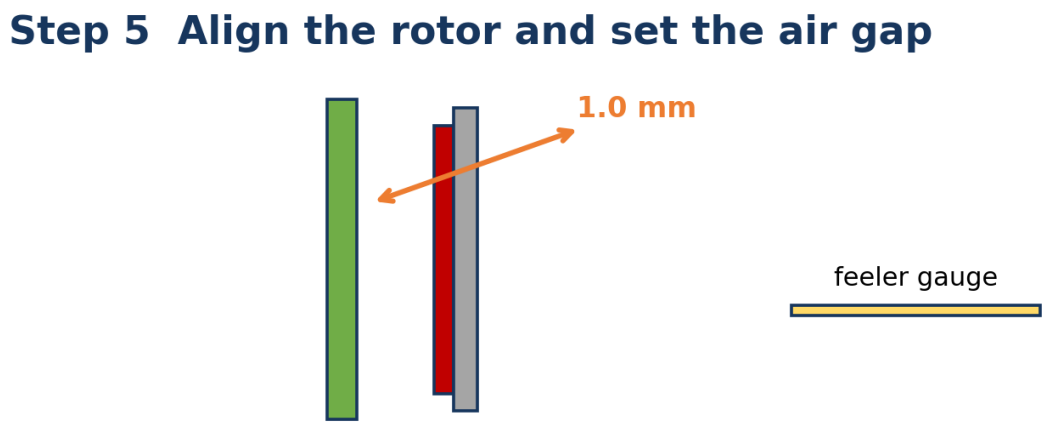



*Figure 6 Step 5 Set alignment and air gap*

Mount the stator and rotor on the rigid base. Set a nominal 1.0 mm face gap with feeler gauges at several points. Rotate through 360 degrees and confirm that the minimum gap remains positive everywhere.

### Step 6 Install sensors and guard

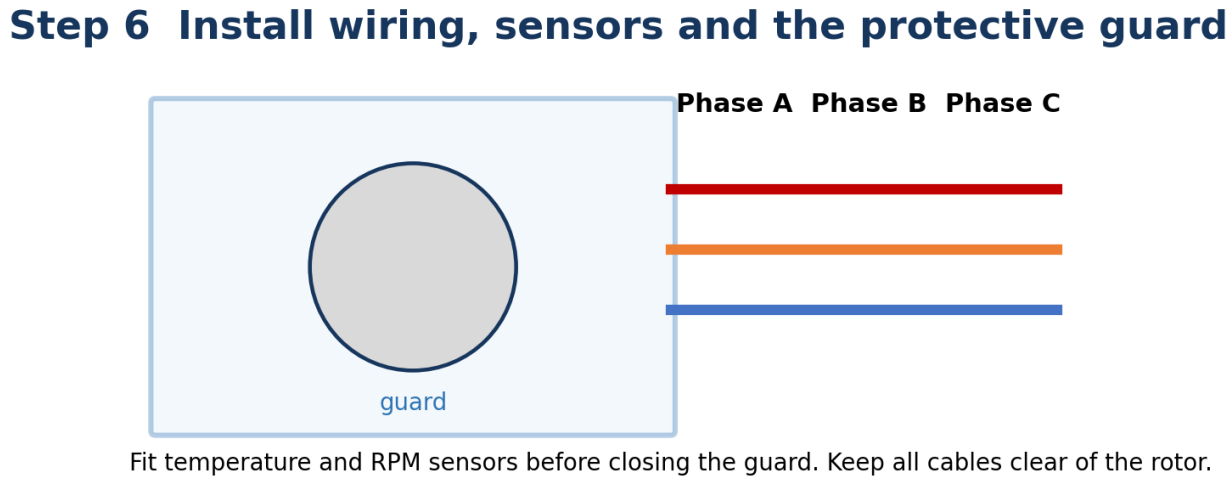



*Figure 7 Step 6 Install sensors and guard*

Route the three phase conductors away from the rotor. Attach thermocouples to the PCB winding region and install the RPM sensor. Fit the clear polycarbonate guard before any powered test.

### Step 7 Connect the test bench

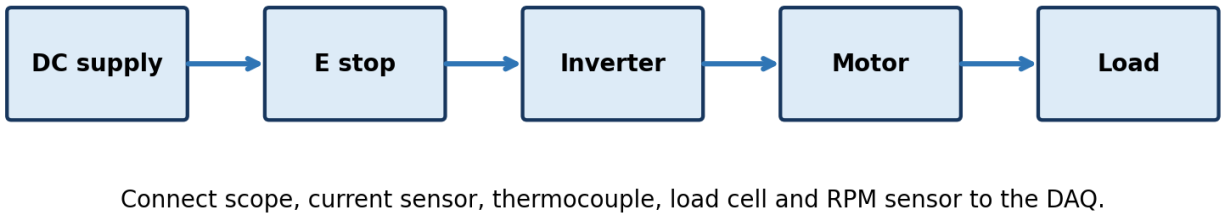



*Figure 8 Step 7 Connect the test bench*

Connect the bench supply through the fuse and latching emergency stop to the inverter. Connect the inverter phases to the stator. Connect voltage, current, speed, torque and temperature instruments to the oscilloscope or DAQ.

### Step 8 Complete the first powered test

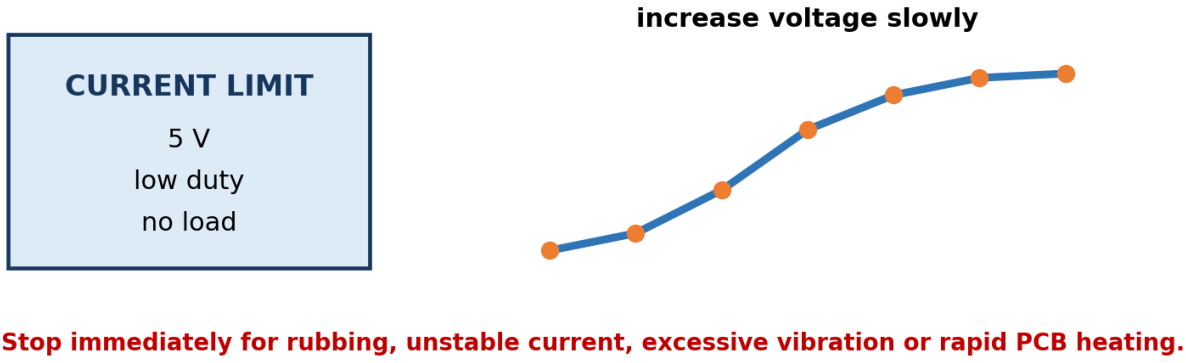



*Figure 9 Step 8 Complete the first powered test*

Set the bench supply current limit low, begin around 5 V and command the lowest practical duty cycle or electrical frequency. Observe phase current, vibration and temperature. Increase only after stable no-load operation is demonstrated.

## Final assembly

The completed Rev0 rig combines the mechanical motor, guarded rotating assembly, current-limited drive and measurement chain. The same base instrumentation can later be reused for the two-stator Rev1 stack.

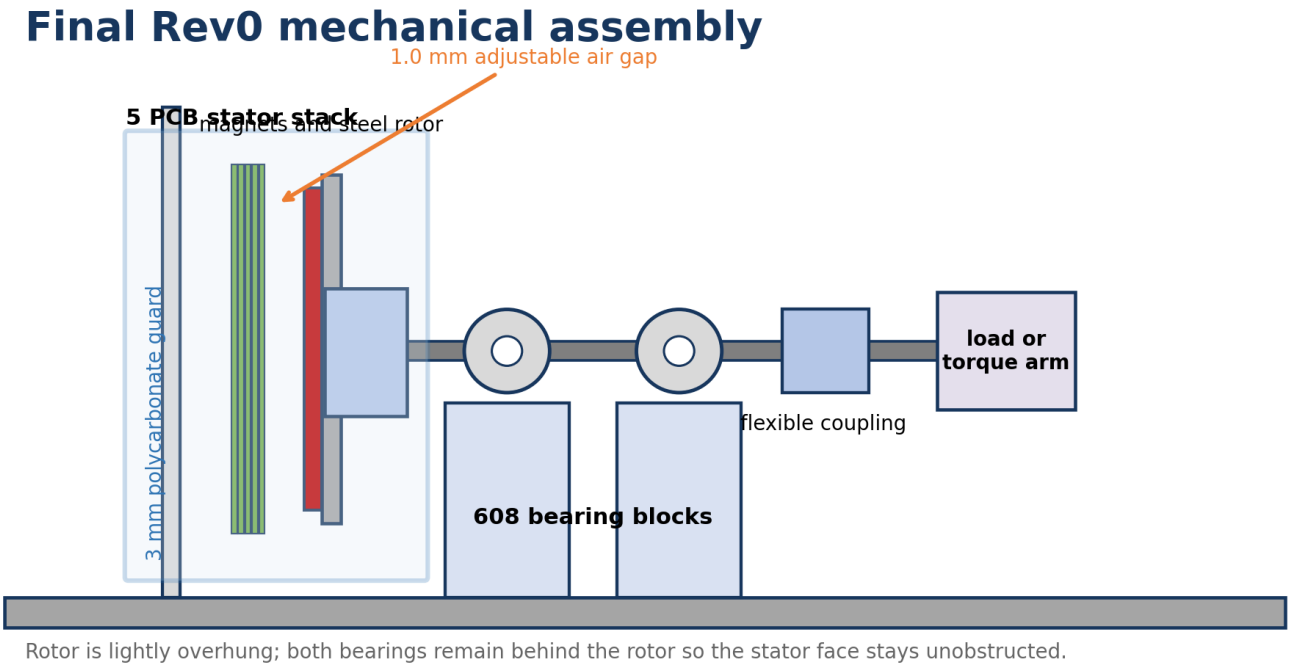



*Figure 10 Final Rev0 mechanical arrangement*

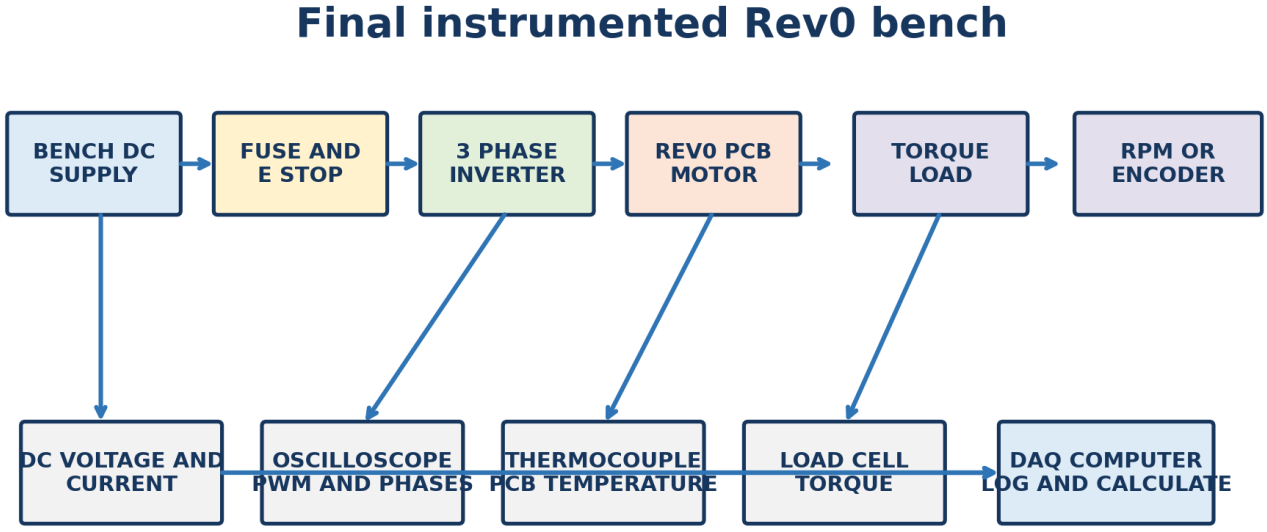



*Figure 11 Final Rev0 drive and measurement connections*

## Pre power sign off

| | | | |
|---|---|---|---|
| **Date** | _____ | **Operator** | _____ |
| **Supervisor** | _____ | **Test reference** | _____ |

- ☐ All 16 magnets are retained and the polarity alternates around the rotor.
- ☐ Rotor bolts, hub clamp and shaft collars are tight and thread-locked where appropriate.
- ☐ Rotor turns freely by hand with no PCB contact through one complete revolution.
- ☐ Minimum air gap is measured and recorded at several angular positions.
- ☐ Phase resistances are similar and no phase is shorted to the frame or rotor.
- ☐ The inverter current limit and bench-supply current limit are configured.
- ☐ Emergency stop interrupts inverter power and is reachable from the operator position.
- ☐ Polycarbonate guard is installed and no person is in the rotor plane.
- ☐ Voltage, current, speed and temperature channels read plausibly before rotation.
- ☐ A test log sheet and shutdown thresholds are ready before energisation.

## Supplier and technical references

The links below are request and sourcing starting points. Confirm stock, final dimensions, current rating, price and delivery before an order is placed.

1. Paul Gould magnetless motor repository
2. JLCPCB
3. UMAGNETS 20 x 6 x 2 mm N35 magnet
4. BMG 608 bearing guide and shop
5. RS South Africa shaft collars
6. Micro Robotics South Africa
7. VESC motor-control platform
8. SimpleFOC motor-driver limitations and specifications
9. Allegro ACS712 current-sensor reference
10. Maizey Plastics
11. Pratley adhesives

### Revision note

This procurement baseline applies to the Rev0 single-rotor, single-stator-stack motor using five 1.0 mm PCBs. Rev1 requires a separate release because the R-S-R-S-R stack introduces three rotor discs, four magnet faces, four air gaps and a different shaft-length and axial-load arrangement.